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Setup and configure a LAN (Local Area Network) using switch and ethernet cable

Aim:

Setup and configure a LAN (local area network) using a switch and ethernet cable in your lab.

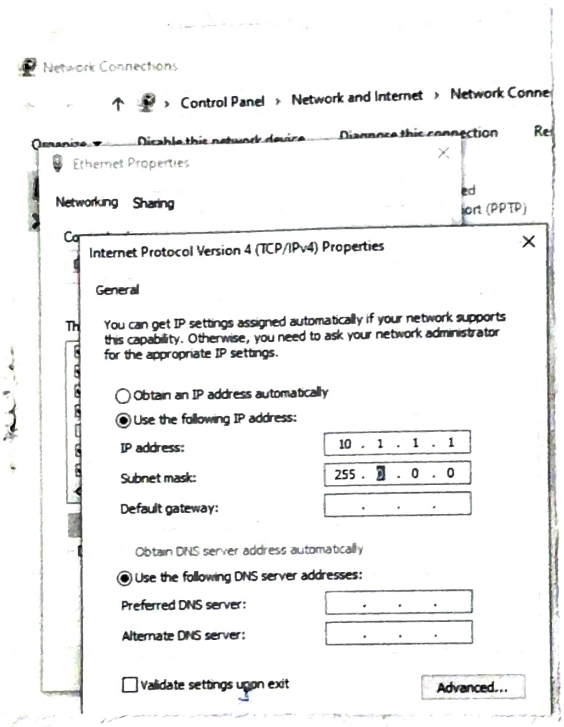
What is LAN:

A local area network (LAN) refers to a network that connects devices within a limited area, such as an office building, school or home. It enables users to share resources, including data, printers and internet access. LAN connects the devices to promote collabration and tranfer information between users, such as computers, printers, servers and switches.

How to set up a LAN:

Step 1:

plan and design an appropicate network topology taking into account network requirements and equipment location


Network Connections
Control Panel › Network and Internet › Network Conne
Ethernet Properties
Networking Sharing
Internet Protocol Version 4 (TCP/IPv4) Properties
General
You can get IP settings assigned automatically if your network supports this capability. Otherwise, you need to ask your network administrator for the appropriate IP settings.
Obtain an IP address automatically
Use the following IP address:
IP address: 10 . 1 . 1 . 1
Subnet mask: 255 . 0 . 0 . 0
Default gateway:
Obtain DNS server address automatically
Use the following DNS server addresses:
Preferred DNS server:
Alternate DNS server:
Validate settings upon exit
Advanced...

Step 2:

you can take 4 computers, a switch with 8, 16 (or) 24 ports which is sufficient for network of these sizes and 4 ethernet cables.

Step 3:

connect your computers to network switch via an ethernet cable. which is an simple as plugging one end of the ethernet cable into your computer and other end into your network switch

Step 4: Assign IP address to your PC's

1. log on to the client computer as admin or as owner
2. click network and internet connection
3. Right click local area connection / Ethernet → Go to properties → select internet protocol (TCP/IP V4) → click on properties → select use the following ip address option and assign ip address.

Step - 5: configure ip network switch

1. connect your computer to the switch
2. log in to the web interface
3. configure the basic settings
4. Assign IP address as: 10.1.1.5 subnet

mask 255.0.0

Step 6:

Check the connectivity between switch and other machine by using ping command. in the command prompt of the device.

Step 7:

Select a folder → go to properties → click sharing tab → show it with everyone on the same LAN

step 8:

Try to access the shared folder from other compuler of the network.

Result:

Thus the setup and configure of LAN using switch and ethernet cable was successfully completed